length of stay shall be such that vibration is reduced to a practical minimum. In cargo holds, ladders shall be designed and arranged so that th e risk of damage from cargo handling gear is minimized.

6 The width of ladders between stringers shall not be less than 400 mm. The treads shall be equally spaced at a distance apart, measured vertically, of between 250 mm and 300 mm. When steel is used, the treads shall be formed of tw o square bars of not less that 22 mm by 22 mm in section, fitted to form a horizontal step with the edges pointing upward. The treads shall be carried through the side stringers and attached thereto by double continuous welding. All sloping ladders shall be provided with handrails of s ubstantial construction on both sides, fitted at a convenient distance above the treads.

7 No free-standing portable ladder shall be more than 5 m long.

8 Portable ladders more than 5 m long may only be utilized if f itted with a remotely controlled mechanical device to secure the upper end of the ladder.

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SOURCE: MSC.133(76).pdf
7 No free-standing portable ladder shall be more than 5 m long.

8 Portable ladders more than 5 m long may only be utilized if f itted with a remotely controlled mechanical device to secure the upper end of the ladder.

9 Movable means of access includes such devices as:

.1 hydraulic arm fitted with a stable base and with local control at the safety cage. The operational conditions should be in accordance with applicable safety requirements of the manufacturer; and

.2 wire lift platform.

10 For bulk carriers, access ladders to a cargo hold shall be:

.1 where the vertical distance between the upper surface of adjacent decks or between deck and the bottom of the cargo space is not more than 6 m, either a vertical ladder or an inclined ladder; and

.2 where the vertical distance between the upper surface of adjacent decks or between deck and the bottom of the cargo space is more than 6 m, an inclined ladder or ladders, except the uppermost 2.5 m of a cargo space measured clear of overhead obstructions and the lowest 6 m may have vertical ladders, provided that the vertical extent of the inclined ladder or ladders connecting the vertical ladders

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SOURCE: MSC.133(76).pdf
overhead obstructions and the lowest 6 m may have vertical ladders, provided that the vertical extent of the inclined ladder or ladders connecting the vertical ladders
is not less than 2.5 m.
RESOLUTION MSC.133(76)
(adopted on 12 December 2002)
ADOPTION OF TECHNICAL PROVISIONS FOR MEANS OF

Table 1 - Means of access for oil tankers

1 Water ballast tanks, except those specified in the right column, and cargo oil tanks	
2 Wing water ballast tanks of less than 5 m width forming double side spaces and their bilge hopper sections	
Access to the overhead structure	
1.1 For tanks of which the height is 6 m and over, permanent means of access shall be provided in accordance with .1 to .3:	
.1 continuous athwartship permanent access arranged at the transverse bulkheads and at every deck transverse, at a minimum of 1.8 m to a maximum of 2.5 m below the overhead structure. If the access is fitted on the side of the unobstructed side of the web plating, then lightening holes of at least 300 mm diameter shall be fitted in the web plating, providing	

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SOURCE: MSC.133(76).pdf

a maximum of 2.5 m below the overhead structure. If the access is fitted on the side of the unobstructed side of the web plating, then lightening holes of at least 300 mm diameter shall be fitted in the web plating, providing access adjacent to both sides of each tripping bracket;	
.2 at least one longitudinal permanent means of access at a minimum of 1.8 m to a maximum of 2.5 m below the overhead structure. Where the longitudinal bulkhead contains attached framing, the access shall be provided at that side; and	
.3 access between the arrangements specified in .1 and .2 and from the main deck to either .1 or .2.	
1.2 For tanks of which the height is less than 6 m, raft or portable means may be utilized in lieu of the permanent means of access.	
2.1 Where the vertical distance between horizontal upper stringer and deck head exceeds 6 m, one continuous permanent means of access shall be provided for the full length of the tank with a means to allow passing through transverse swash bulkheads installed a	

minimum of 1.8 m to a maximum of 2.5 m from the overhead structure with a vertical access ladder at each end and mid-span of tank.

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SOURCE: MSC.133(76).pdf
for the full length of the tank with a means to allow passing through transverse swash bulkheads installed a minimum of 1.8 m to a maximum of 2.5 m from the overhead structure with a vertical access ladder at each end and mid-span of tank.
2.2 For bilge hopper sections of which the vertical distance from baseline to the upper knuckle point is 6 m and over, one longitudinal permanent means of access shall be provided for the full length of the tank. It shall be accessible by vertical permanent means of access at both ends of the tank.
2.3 Where the vertical distance referred to in 2.2 is less than 6 m, portable means of access may be utilised in lieu of the permanent means of access. To facilitate the operation of the portable means of access, in-line openings in horizontal stringers should be provided. The openings should be of an adequate diameter and should have suitable protective railings.

2.4 Whenever practicable, the distance between the overhead structure and the uppermost longitudinal stringer and between the longitudinal stringers should not exceed 6 m.
Access to the vertical structures
1.3 For tanks of which the height is 6 m and over,

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SOURCE: MSC.133(76).pdf
overhead structure and the uppermost longitudinal stringer and between the longitudinal stringers should not exceed 6 m.
Access to the vertical structures
1.3 For tanks of which the height is 6 m and over, containing internal structures, permanent means of access shall be provided to each transverse web.

1.4 For tanks of which the height is less than 6 m, raft or portable means may be utilized in lieu of the permanent means of access.

2.5 Vertical permanent means of access shall be provided to each transverse web in the following cases where the vertical distance is 6 m and over:

.1 from baseline to the upper knuckle point of the bilge hopper section;

.2 from the upper knuckle point of the bilge hopper section to main deck where no horizontal stringers are provided; and

.3 between horizontal stringers.

2.6 Access holes within 600 mm of the stringer shall be provided in each transverse web/swash bulkhead above each stringer and tank base.

2.7 In the case where the vertical distance referred to in 2.5 is less than 6 m, portable means may be utilised in lieu of the permanent means of access.
RESOLUTION MSC.133(76)

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SOURCE: MSC.133(76).pdf
above each stringer and tank base.

2.7 In the case where the vertical distance referred to in 2.5 is less than 6 m, portable means may be utilised